

Zuber Data Analysis Project

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Project Overview

Zuber is a new ride-sharing company preparing to launch operations in Chicago. As a data analyst, my task was to explore available data to uncover insights about passenger preferences and the impact of external factors—particularly weather—on ride patterns. The analysis used data from several tables containing information on taxi rides, vehicles, neighborhoods, and weather conditions.

Data Description

The database used in this project included the following tables:

- neighborhoods – Contains information about Chicago neighborhoods (neighborhood_id, name).
- cabs – Contains data on individual vehicles (cab_id, vehicle_id, company_name).
- trips – Includes ride-level details (trip_id, cab_id, start_ts, end_ts, duration_seconds, distance_miles, pickup_location_id, dropoff_location_id).
- weather_records – Provides hourly weather data (record_id, ts, temperature, description).

Although the trips and weather_records tables were not directly connected, they could be linked through the timestamp fields (trips.start_ts and weather_records.ts) to analyze how weather affected ride activity.

1. Ride Volume by Taxi Company (November 15–16, 2017)

I calculated the total number of rides for each taxi company during November 15–16, 2017. Results were sorted in descending order to identify the most active companies.

```
SELECT
    cabs.company_name,
    COUNT(trips.trip_id) AS trips_amount
FROM
    cabs
INNER JOIN trips ON trips.cab_id = cabs.cab_id
WHERE
    trips.start_ts::date BETWEEN '2017-11-15' AND '2017-11-16'
GROUP BY
    cabs.company_name
ORDER BY
    trips_amount DESC;
```

2. Ride Counts for 'Yellow' and 'Blue' Companies (November 1–7, 2017)

I examined the number of rides completed by companies whose names included the words “ Yellow ” or “ Blue ” during November 1-7, 2017, to compare their market performance.

```
SELECT
    cabs.company_name AS company_name,
    COUNT(trips.trip_id) AS trips_amount
FROM
    cabs
INNER JOIN trips ON trips.cab_id = cabs.cab_id
WHERE
    CAST(trips.start_ts AS date) BETWEEN '2017-11-01' AND '2017-11-07'
    AND cabs.company_name LIKE '%Yellow%'
GROUP BY
    company_name
UNION ALL
SELECT
    cabs.company_name AS company_name,
    COUNT(trips.trip_id) AS trips_amount
FROM
    cabs
INNER JOIN trips ON trips.cab_id = cabs.cab_id
WHERE
    CAST(trips.start_ts AS date) BETWEEN '2017-11-01' AND '2017-11-07'
    AND cabs.company_name LIKE '%Blue%'
GROUP BY
    company_name;
```

3. Comparing Top Companies vs. Others

I grouped the ride data for November 1-7, 2017, into three categories: Flash Cab, Taxi Affiliation Services, and Other. This confirmed that the first two companies dominated the market.

```
SELECT
    CASE
        WHEN company_name = 'Flash Cab' THEN 'Flash Cab'
        WHEN company_name = 'Taxi Affiliation Services' THEN 'Taxi Affiliation Services'
        ELSE 'Other'
    END AS company,
    COUNT(trips.trip_id) AS trips_amount
FROM
    cabs
INNER JOIN trips ON cabs.cab_id = trips.cab_id
WHERE
    CAST(trips.start_ts AS date) BETWEEN '2017-11-01' AND '2017-11-07'
GROUP BY
    company
ORDER BY
    trips_amount DESC;
```

4. Neighborhood Identification

To analyze rides between the Loop and O ' Hare International Airport, I retrieved their respective neighborhood identifiers.

```
SELECT
    neighborhood_id,
    name
FROM
    neighborhoods
WHERE
    name LIKE 'Loop'
    OR name LIKE '%Hare';
```

5. Weather Condition Categorization

Each hourly weather record was classified as “ Good ” or “ Bad ” depending on whether the description included the words rain or storm.

```
SELECT
    ts,
    CASE
        WHEN description LIKE '%rain%' OR description LIKE '%storm%' THEN 'Bad'
        ELSE 'Good'
    END AS weather_conditions
FROM
    weather_records;
```

6. Ride Duration Analysis: Loop → O ' Hare on Saturdays

Finally, I examined how ride durations between the Loop (pickup_location_id = 50) and O ' Hare (dropoff_location_id = 63) changed depending on weather conditions on Saturdays.

```
SELECT
    trips.start_ts AS start_ts,
    CASE
        WHEN weather_records.description LIKE '%rain%' THEN 'Bad'
        WHEN weather_records.description LIKE '%storm%' THEN 'Bad'
        ELSE 'Good'
    END AS weather_conditions,
    trips.duration_seconds AS duration_seconds
FROM
    trips
INNER JOIN weather_records ON weather_records.ts = trips.start_ts
WHERE
    trips.pickup_location_id = '50'
    AND trips.dropoff_location_id = '63'
    AND EXTRACT(DOW FROM trips.start_ts) = 6
ORDER BY
    trip_id;
```

Conclusion

Through SQL-based exploratory data analysis, I identified the leading taxi companies, assessed ride distribution among competitors, and evaluated the influence of weather on travel times between major Chicago neighborhoods. The findings suggest that both company activity and environmental conditions significantly affect trip patterns—valuable insights for Zuber ’ s launch strategy and operational planning.